Q-volution Hackathon 2026

TEAM
Quantum Pulse

Energy Grid Optimization via Quantum Decomposition

A Hybrid QAOA Pipeline Validated on the Rigetti Ankaa-3

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The hackathon asks us to find the Maximum Power Energy Section (MPES) of a South Carolina transmission network [1, 2]. MPES is a weighted MaxCut problem. We have two instances: **Problem A** (21 nodes, 28 edges) is small enough to solve exactly and verify against. **Problem B** (180 nodes, 226 edges) is the real challenge. Running QAOA directly on 180 qubits is not possible on any hardware today. We built a hardware-aware hybrid pipeline that makes the 180-node problem executable within current NISQ constraints. We refer to this approach as *Hybrid Quantum-Classical Decomposition under Hardware Constraints*: a methodology that partitions large optimization problems into hardware-feasible quantum subproblems, executes them under explicit noise budgets, and reconstructs a global solution via classical refinement.

Problem and Formulation

Each instance is a weighted graph $G = (V, E, w)$ where nodes are buses, edges are transmission lines, and w_{ij} is the admittance of each line. The objective is

$$\max_{z\in\{-1,+1\}^n} \sum_{(i,j)\in E} \tfrac{w_{ij}}{2}(1 - z_i z_j).$$

For QAOA this becomes the Ising cost Hamiltonian $C = \sum_{(i,j)\in E} \frac{w_{ij}}{2}(1 - Z_i Z_j)$, and $\langle\psi|C|\psi\rangle$ returns the cut value directly. Both graphs are sparse and connected. Problem A has density $\rho = 0.133$ and Fiedler value $\lambda_1 = 4.87$; Problem B has $\rho = 0.014$ and $\lambda_1 = 0.105$. That small Fiedler value signals weak global cohesion and natural internal structure — exactly the property that makes decomposition viable here.

Why Direct QAOA Fails at This Scale

Standard depth-p QAOA builds the state

$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k B}\, e^{-i\gamma_k C}\, |+\rangle^{\otimes n},$$

placing one R_{ZZ} gate per edge per layer. At $p = 1$, Problem B needs 226 two-qubit gates before routing. Spectral embedding onto the Ankaa-3's 10×9 square-grid coupler layout pushes this to 244 native iSWAP gates, and the device only has 84 physical qubits — it cannot hold 180 logical ones. Classical statevector simulation would require approximately 2.45×10^{55} bytes, which is infeasible on any existing classical infrastructure.

The gate budget is just as limiting. At per-gate fidelity $F_{2Q} = 0.990$, circuit fidelity decays as F_{2Q}^k. The maximum useful gate count before fidelity falls below 0.50 is

$$k_{\max} = \left\lfloor \frac{\ln 0.5}{\ln 0.990} \right\rfloor = 68.$$

Problem B needs 244 embedded gates at $p = 1$; predicted fidelity is 0.086. Every measurement would be pure noise.

Hardware-Aware Decomposition Pipeline

We cut Problem B into 16 clusters, run QAOA on each, stitch the results into a global partition, and repair boundaries classically. Every stage was designed around the hardware's actual numbers.

Cluster construction. BFS partitioning with hard constraints: at most 12 nodes and 15 internal edges per cluster, keeping native iSWAP loads between 18 and 28. All 16 clusters land at 11–12 nodes (mean 11.25) with 9–14 internal edges (mean 10.75). Boundary edges are excluded from each subproblem's QAOA — the structural cost of decomposition, recovered later.

Survival model. Using measured Ankaa-3 device parameters — median iSWAP fidelity $\bar{F}_{2Q} = 0.9852$, median $T_2 = 22.08\,\mu$s — we pre-compute a gate-plus-decoherence survival probability for every cluster before touching the hardware:

$$S(N_{2Q}) = \bar{F}_{2Q}^{N_{2Q}} \cdot \exp\!\left(-\tfrac{N_{2Q}\cdot t_{2Q}}{T_2}\right), \quad t_{2Q} = 200\,\text{ns}.$$

All 16 clusters predict $S \in [0.512, 0.650]$, every one above the collapse threshold $S = 0.20$, defined as the regime where signal-to-noise becomes indistinguishable from uniform sampling under 2000-shot measurement. A naive node-count-only partition would push iSWAP loads to 26–38 and minimum predicted survival to 0.403 — dangerously close to threshold.

Cluster QAOA. Each subgraph $G_c = G_B[\mathcal{V}_c]$ is solved at $p = 1$. Parameters (γ, β) are found by coarse 2D grid search then COBYLA refinement with three restarts, sampling 1024 shots. Exact brute-force MaxCut is also run on each cluster as a quality check.

Stitching and repair. Combining 16 most-probable bitstrings gives raw cut $C_{\text{stitched}} = 5346.62$ (ratio 0.789 vs. greedy). The gap comes from boundary edges QAOA never saw. A one-exchange sweep flips any node v where

$$\Delta_v = \sum_{u\in\mathcal{N}(v)} w_{vu} \cdot \mathbf{1}[s_v = s_u] > 0,$$

adding $\Delta = 1088.98$ to reach $C_{\text{refined}} = 6435.60$ (ratio 0.949).

Hardware Execution on Rigetti Ankaa-3

Three clusters were pre-selected before any hardware job ran: clusters 11, 6, and 5 at the 25th, 50th, and 75th percentiles of native iSWAP load. Angles (γ^*, β^*) from simulation were frozen and transferred unchanged to hardware.

Compilation audit. Every circuit was compiled via QCS and native gate counts were inspected before execution. Clusters 11

and 6 compiled to 9 and 12 native iSWAP gates. Cluster 5 was not: naive qubit mapping returned 60 native iSWAP gates against an expected 18 — a 3.33× overhead, placing predicted survival at $S \approx 0.098$, deep in the collapse regime. We caught this before spending any QPU time.

Embedding refinement. Degree-priority remapping: logical nodes sorted by internal degree are mapped first to well-connected central physical qubits, then multiple randomised orderings are compiled and the minimum-gate result is selected. Cluster 5 dropped from 60 to 30 compiled iSWAP gates (predicted survival 0.488). Final spread: 9, 12, and 30 compiled gates across the feasibility envelope.

QPU results (2000 shots per cluster on the physical Ankaa-3):

Cluster	iSWAP	Ideal	HW cut	Pred.	Meas.
Cluster 11	9	427.57	322.35	0.806	0.754
Cluster 6	12	375.84	150.08	0.750	0.399
Cluster 5	30	295.46	122.88	0.488	0.416

Clusters 11 and 5 land close to predictions (errors −0.052 and −0.072), validating the survival model at both shallow and deep gate counts. Cluster 6 is the outlier: measured ratio 0.399 vs. predicted 0.750, error −0.351. Twelve iSWAP gates alone are unlikely to explain this deviation under the median fidelity model. The likely cause is per-qubit noise on the specific physical qubits assigned — something the chip-median model cannot predict. A readout calibration attempt for Cluster 6 timed out at latency > 30,000 ms during the session, consistent with elevated QPU load or a degraded qubit in that partition.

Post-compilation embedding refinement. To isolate the effect of qubit placement from algorithmic changes, we re-executed selected clusters after applying degree-priority embedding refinement. No angles were re-optimized; (γ^*, β^*) were kept frozen. The only change was the logical-to-physical qubit mapping selected via compilation audit.

Cluster	iSWAP	HW cut (baseline)	HW cut (refined)	Runtime (s)
Cluster 11	9	322.35	359.80	6.6
Cluster 5	30	122.88	176.25	6.62

Both clusters improved under refined embedding without any change to circuit depth or parameters. Cluster 11 increased from 322.35 to 359.80 (+11.6%), while Cluster 5 increased from 122.88 to 176.25 (+43%). Since logical depth and shot count were unchanged, the improvement is attributable purely to compilation-aware qubit placement. This confirms that native-gate topology and qubit-specific noise dominate performance in this NISQ regime.

Parameter stability was confirmed by a γ-sweep for Cluster 11 (7 points, 1500 shots each): best cut 371.11 at $\gamma = 2.796$, close to the primary run at 322.35. A 5×5 (γ, β) scan at 1000 shots returned its best result at the frozen angles (2.996, 0.447). A shot-scaling study (500–4000 shots) found no monotonic improvement, consistent with a distribution concentrating around a small set of high-cut bitstrings.

Problem A: Exact Verification

Problem A was solved by exhaustive enumeration over all $2^{21} = 2{,}097{,}152$ partitions: maximum cut 3728.41 at bipartition $\{1, 2, 4, 8, 10, 11, 14, 15, 16, 19\}$ vs. $\{0, 3, 5, 6, 7, 9, 12, 13, 17, 18, 20\}$. Greedy one-exchange reaches 3632.36 (ratio 0.974). An 8-node hardware subgraph centred on node 1 (highest weighted degree, 736.05) gives exact cut 1165.84; greedy matches exactly (ratio 1.000), setting the Ankaa-3 QAOA target.

Results and Discussion

Method	Cut value	Ratio
Random (best of 20)	4405.41	0.650
QAOA stitched, ideal	5346.62	0.789
QAOA stitched, noise model	3602.92	0.531
Simulated Annealing (20k iter.)	6591.01	0.972
QAOA + one-exchange refinement	6435.60	0.949
Greedy one-exchange (baseline)	6778.19	1.000

Clustered QAOA alone does not beat greedy, and hardware noise reduces the raw signal further. The full hybrid pipeline — quantum decomposition followed by one-exchange repair — reaches 94.9% of greedy on a 180-node problem actually executed on a real superconducting processor. No quantum advantage is claimed; that was never the target.

What this project demonstrates is that it is possible to take a real-world 180-node optimisation problem, engineer a hardware-safe decomposition, run the pieces on a QPU, and recover a useful global solution. The chain that made it work: Fiedler-guided decomposability check, hardware-constrained partitioning, pre-registered survival predictions, compilation audit, degree-priority embedding refinement, frozen-angle QPU execution, and lightweight classical repair. Every stage was verified against real numbers before the hardware job ran. The Cluster 5 near-miss — 60 compiled iSWAP gates before refinement, 30 after — shows concretely why the compilation audit is not optional. That engineering discipline is as much a contribution as the quantum results themselves.

References

[1] J. Stubbemann, *Energy Grid Optimization – Problem A*, ver. 0.0.0, Aqora, 2026. DOI: 10.82503/aqora.2602.PQ7M5. License: CC-BY-SA-4.0.

[2] J. Stubbemann, *Energy Grid Optimization – Problem B*, ver. 0.0.0, Aqora, 2026. DOI: 10.82503/aqora.2602.7H6J5. License: CC-BY-SA-4.0.